

1. Introduction & Theoretical Physics Foundation

1.1 The Failure of Deterministic Binary Invariant Pipelines

Modern silicon-based computation remains shackled to the legacy of the Von Neumann architecture, an operational model that mandates absolute, zero-defect state determinism across a polarized binary grid ($B \in \{0, 1\}$). While mathematically convenient for linear operations, this convention imposes an unsustainable thermodynamic penalty when interacting with real-world, stochastic data domains.

To force a high-variance, continuous analog wavefront into discrete binary states, standard architectures require immense thermal dissipation and massive clock-cycle overhead to continually correct for noise, jitter, and logical drift. When an anomaly, structural hallucination, or adversarial payload penetrates this rigid perimeter, the system either executes the corrupted packet blindly or stalls completely, inducing thread blockages and memory leaks.

Nature, by contrast, rejects static binary determinism. Biological systems and quantum electrodynamic fields operate on a principle of *Dynamic Equilibrium through Perpetual Unachievable Balance*. Physical systems are in a state of continuous contraction, expansion, decay, and transformation—constantly migrating toward an ideal energetic equilibrium they can never permanently settle into.

By modeling this natural imbalance as an active operational feature rather than a system defect, we can construct a computational framework that treats information as a fluid wave geometry. This paper proposes the **Topological Sieve Gate**, a hardware-emulation processing layer that replaces traditional linear "Fetch, Decode, Execute" pipelines with multi-pass **Geometric State Refraction**.

Operating natively on balanced ternary space boundaries ($\Sigma T \in \{-1, 0, +1\}$), this architecture absorbs data gaps as an active state of **Unknown / Shortfall (\$0.05)** and structural irrelevance as a **Void / Path Target (-1.0)**, allowing toxic or non-harmonic strings to be shed at the outer boundary shell edge without consuming downstream processor logic resources.

1.2 Mathematical Derivation of the Sieve Threshold (θ)

To establish an unalterable spatial filter capable of distinguishing harmonized signal payloads from chaotic entropic noise, the boundary grid must be mapped to a permanent geometric ratio. We define the global **Sieve Threshold (θ)** as the product of circular spatial closure, asymmetric evolutionary growth scaling, and trinary spatial dimensionality split:

$$\theta = \pi \cdot \phi \cdot \left(\frac{1}{3}\right)$$

Where:

- π represents the transcendental constant of circular phase closure (≈ 3.14159265359), ensuring geometric coherence across cyclical wave rotations.

- ϕ represents the continuous optimization scale of the Golden Ratio (≈ 1.61803398875), derived from the positive root of the algebraic equation:

$$\phi^2 - \phi - 1 = 0 \Rightarrow \phi = (1 + \sqrt{5}) / 2$$

The inclusion of ϕ enforces an optimal, non-periodic structural scaling factor that prevents the wave vector from entering destructive harmonic resonance loop locks with external background noise.

- $(1/3)$ represents the spatial refraction index coefficient, splitting the continuous wavefront symmetrically across the three foundational dimensions of our balanced ternary state space.

Evaluating this composite geometric limit yields the definitive structural refraction angle:

$$\theta = (\pi \cdot 1.61803398875) / 3 \approx 1.6944062 \text{ radians} \quad (\approx 97.082^\circ)$$

1.3 Geometric Propagation and Spatial Refraction Proof

Let the processing fabric be defined as a continuous complex wavefront state vector Ψ operating within a 14-node coordination register matrix. When an external data stream is compressed into balanced ternary space, it materializes as an initial spatial vector state $\Psi_0 \in \mathbb{R}^{14}$.

To simulate physical wave propagation through a refractive boundary medium (such as a 3D Hexagonal Close-Packed crystalline lattice), the vector undergoes an index translation cycle. The transformation is driven by a discrete spatial shift operator matrix σ , which executes a single-node circular roll to model physical movement across adjacent register paths:

$$\sigma \Psi_n = \text{roll}(\Psi_n, 1)$$

Because the medium splits energy evenly across the ternary plane, the rolled matrix is scaled by the $1/3$ physical refraction index, establishing the spatial refraction rotation matrix Ω_n :

$$\Omega_n = \sigma \Psi_n \cdot (1/3)$$

The continuous wavefront evolution from state Ψ_n to the subsequent state Ψ_{n+1} is computed by projecting the vector across the transcendental orthogonal bounds of the sieve threshold angle θ :

$$\Psi_{n+1} = \Psi_n \cos(\theta) + \Omega_n \sin(\theta)$$